Python 3.11 | Built with Streamlit | Status Completed | License MIT | last commit last wednesday

SpaceY Launch Prediction Dashboard



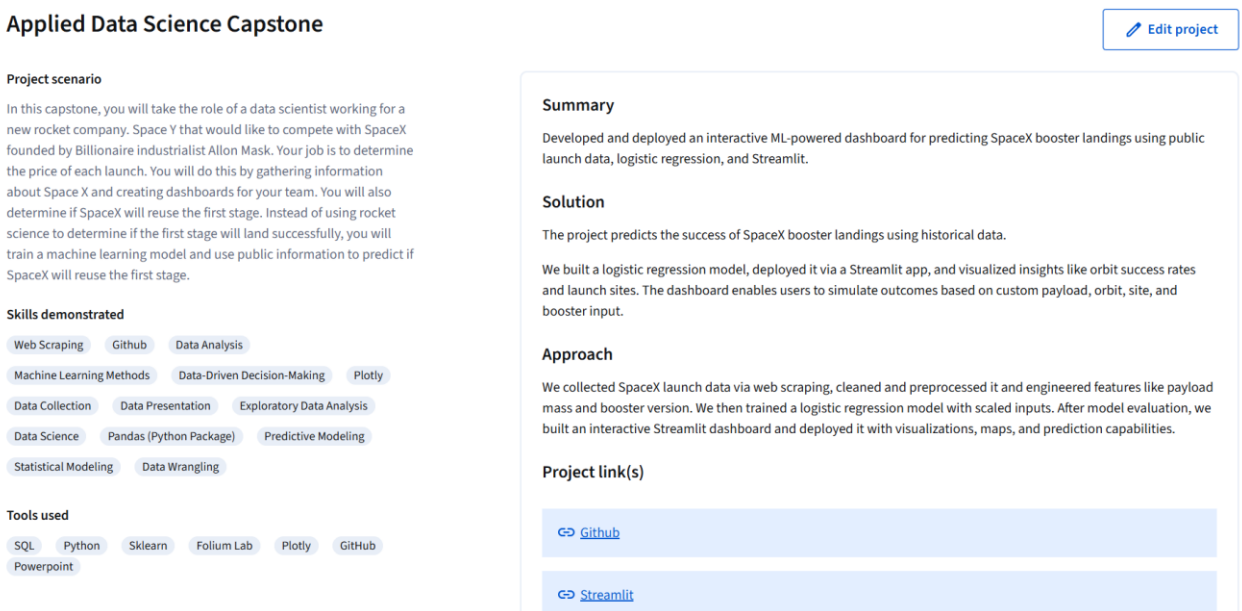
An interactive machine learning-powered web dashboard that predicts the success of SpaceX booster landings using public data, logistic regression, and modern data visualization tools.

Live App: Click here to visit
Repo: GitHub Source Code

Project Overview

This capstone project simulates the role of a **data scientist at SpaceY**, a fictional competitor to SpaceX. Using publicly available SpaceX launch data, we:

- Predict whether the **booster will land successfully**
- Visualize success rates across **orbits** and **launch sites**
- Provide an interactive tool for engineers and stakeholders



Features

Feature	Description
Launch Simulation Inputs	Select payload mass, orbit type, booster version, and launch site
Booster Landing Prediction	ML model predicts whether the booster will land
Prediction Confidence	Displays probability/confidence in prediction
Launch Site Map	Interactive Folium map showing launch locations
Orbit vs Success Rate	Plotly visualization of orbit-wise success trends

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Tech Stack

- **Python 3.10**
- **Pandas, NumPy** – Data wrangling
- **Scikit-learn** – Model training (Logistic Regression)
- **Joblib** – Model serialization
- **Plotly** – Interactive orbit
- visualizations **Folium + streamlit-**
- **folium** – Maps **Streamlit** –
  Dashboard frontend & backend

Project Structure

spacey_capstone/

```
├── dashboard/
│   ├── app.py
│   ├── model.pkl
│   └── scaler.pkl
├──data/
│   ├──cleaned_spacex_launches.csv
├──notebooks/
│   ├── SpaceY_Modeling.ipynb
├── requirements.txt
├──README.md
```

How It Works

1. Load historical SpaceX launch data
2. Clean and one-hot encode categorical variables
3. Train a **logistic regression** model to predict success (1 = landed, 0 = failed)
4. Build a **Streamlit dashboard** to accept launch parameters as user input
5. Predict in real-time and display result + visualizations

Installation & Local Use

```
git clone https://github.com/your-username/spacey-
dashboard.git cd spacey-dashboard
pip install -r requirements.txt
streamlit run dashboard/app.py
```

```
<style type="text/css">@media print {
    *, :after, :before {background: 0 0 !important;color: #000 !important;box-shadow: none !important;text-
    shadow: none !im a, a:visited {text-decoration: underline}
    a[href]:after {content: " (" attr(href) ")"}
    abbr[title]:after {content: " (" attr(title) ")"}
    a[href^="#"]:after, a[href^="javascript:"]:after {content:
    ""} blockquote, pre {border: 1px solid #999;page-break-
    inside: avoid} thead {display: table-header-group}
    img, tr {page-break-inside:
    avoid} img {max-width: 100%
    !important} h2, h3, p
    {orphans: 3;widows: 3} h2, h3
    {page-break-after: avoid}
}
html {font-size: 12px}
@media screen and (min-width: 32rem) and (max-width:
    48rem) { html {font-size: 15px}
}
@media screen and (min-width: 48rem)
    { html {font-size: 16px}
}
body {line-height: 1.85}
.air-p, p {font-size: 1rem;margin-bottom: 1.3rem}
.air-h1, .air-h2, .air-h3, .air-h4, h1, h2, h3, h4 {margin: 1.414rem 0 .5rem;font-weight: inherit;line-
height: 1.42}
.air-h1, h1 {margin-top: 0;font-size: 3.998rem}
.air-h2, h2 {font-size: 2.827rem}
.air-h3, h3 {font-size: 1.999rem}
.air-h4, h4 {font-size: 1.414rem}
.air-h5, h5 {font-size: 1.121rem}
.air-h6, h6 {font-size: .88rem}
.air-small, small {font-size: .707em}
canvas, iframe, img, select, svg, textarea, video {max-width: 100%}
body {color: #444;font-family: 'Open Sans', Helvetica, sans-serif;font-weight: 300;margin: 0;text-
align: center} img {border-radius: 50%;height: 200px;margin: 0 auto;width: 200px}
a, a:visited {color: #3498db}
a:active, a:focus, a:hover {color: #2980b9}
pre {background-color: #fafafa;padding: 1rem;text-align: left}
blockquote {margin: 0;border-left: 5px solid #7a7a7a;font-style: italic;padding: 1.33em;text-
align: left} li, ol, ul {text-align: left}
p {color: #777}</style>
```